



Mattia Simonutti

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About Me

Passionate about artificial intelligence and data science, currently pursuing a Master's degree in Data Science and Artificial Intelligence. Interested in machine learning, deep learning, and data-driven problem solving, with a strong motivation to continuously learn and take on new challenges.

Education

M.Sc. in Data Science and Artificial Intelligence 2024 - Present
University of Trieste

- **Relevant coursework:** , High Performance Computing, Machine Learning Operations, Probabilistic Machine Learning, Statistical Methods, Optimization for Artificial Intelligence, Deep Learning.

B.Sc. in Artificial Intelligence and Data Analytics 2021 - 2024
University of Trieste – Graduated in October 2024, 105/110

- **Most important subjects:** Introduction to basic Machine Learning models and frameworks, Statistical Inference, Data Analysis and construction of predictive models, Introduction to Artificial Intelligence, Numerical Analysis, Advanced and Parallel Programming (C and Python).

- **Thesis:** “*Conformal Inference as Regularization Term in Regression Models*”

Projects

StrokePrediction: Machine Learning models for stroke prediction (Python)

Developed and compared multiple ML models (Neural Networks, Random Forest, SVM, KNN, Logistic Regression) to predict stroke risk, improving model selection through data-driven evaluation. Delivered actionable insights via visualizations and reports, strengthening team decision-making.

Health Care demand Australia (R)

Modeled healthcare demand using Poisson GLMs, GAMs, and Zero-Inflated approaches, enabling accurate representation of complex usage patterns and supporting data-informed healthcare planning.

Evolving Differential Equations with Genetic Programming (Python)

Designed a symbolic regression pipeline to recover governing differential equations from data, demonstrating how data-driven methods can uncover underlying system dynamics.

Diffusion for Molecule Generation (Python)

Built a score-based diffusion model with EGNN architecture for 3D molecule generation, contributing to drug discovery by addressing key challenges in molecular validity and structure reconstruction..

Graph Neural Network and LSTM model for Alzheimer's classification

Developed deep learning models (GNN, GAT, LSTM) for early Alzheimer's detection from MRI data, improving classification performance and supporting early diagnosis through advanced pattern recognition.

Safe and Verified AI: Demographic Fairness for Loan Approval

Implementation and evaluation of a shielded reinforcement learning agent operating in a biased environment, fixing its over-rejection tendency towards a specific demographic group.

Skills

- **Technical:** Python (and related libraries used in projects, such as Pandas, NumPy, Scikit-learn, PyTorch), R, C, SQL (basic), MATLAB and tools such as Git and Github.

Languages

Italian (native), English (independent user).